Felipe Cesário Laterza Lopes

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SUMMARY

Ph.D. student in Physics at Universidade Estadual de Campinas (UNICAMP), in Brazil, and a member of the **LHCb Collaboration** at European Organization for Nuclear Research (CERN), currently based at CERN. Holds a B.Sc. in Physics from UNICAMP. Research experience in **CP violation** in neutrino oscillations and in **charm meson decays**. Current research is an amplitude analysis of the $D^+ \to K^-K^+\pi^+$ decay in LHCb. As a member of the collaboration, affiliated through Centro Brasileiro de Pesquisas Físicas (CBPF), contributes to the Real-Time Analysis (RTA) project.

EDUCATION AND EXPERIENCE

03.2026 - present: **Research Internship Abroad** at **CERN**

Project: Real Time Analysis: the π^0 Calibration and IntelliRTA at LHCb.

Summary: **Software** development for the **trigger** system and for its **monitoring** at the LHCb Experiment, including the deployment of a monitoring interface for the calibration of the Electromagnetic Calorimeter.

Supervisor: Dr. Sascha Stahl.

Financing: FAPESP BEPE Fellowship.

08.2024 - present: **PhD student in Physics** at **UNICAMP**

Thesis: Amplitude Analysis of the $D^+ \to K^-K^+\pi^+$ Decay.

Summary: Development of a **phenomenological model** for the amplitude of the $D^+ \to K^-K^+\pi^+$ decay and implementation of the model for the experimental **amplitude analysis** using data from the **LHCb Experiment** collected during Run 2 of the LHC.

Supervisor: Prof. Dr. Patricia Camargo Magalhaes.

Financing: FAPESP Doctorate Fellowship.

03.2020 - 07.2024: **Bachelor in Physics** at **UNICAMP**

Conclusion Project: Study of CP Violation in Charm Decays.

Summary: Study of **CP Violation** (CPV) **mechanisms** in hadronic two-body decays of the **$\mathbf{D^0}$ meson** ($D^0 \to K^-K^+$ and $D^0 \to \pi^-\pi^+$) and comparison with CPV mechanisms in neutrino oscillations.

Supervisor: Prof. Dr. Patricia Camargo Magalhaes.

Complementary Education

CERN Latin-American **School of High Energy Physics** - CERN, 2025 (64h).
XXIII Jorge André Swieca Summer **School of Particles and Fields** - CBPF, 2025 (40h).
INCT-CERN Brazil **School on Data Analysis** - IFT-UNESP, 2024 (40h).
ICTP-SAIFR/ExoHad **School on Few-Body Physics** - ICTP-SAIFR, 2024 (30h).

International **School on Collider Physics** - Online (UFRN), 2024 (10h).

Professional Experience

03.2026 - present: Research Internship, CERN - Software development for the trigger system.

08.2024 - present: Ph.D. Student, UNICAMP - Full-time research and LHCb member.

02.2024 - 07.2024: **Internship** at Linear Softwares Matematicos Ltda. - Modelling and Inovation Group: back-end software development of mathematical models for optimization.

Research Projects

2026 - present: Real Time Analysis: the π^0 Calibration and IntelliRTA at LHCb

2024 - present: Amplitude Analysis of the $D^+ \to K^- K^+ \pi^+$ Decay

2023: Study of CP Violation in Charm Decays

2022 - 2023: Approach to Neutrino Physics through the DUNE Experiment

SKILLS

Languages

Portuguese - Native;

English - Proficient (Cambridge Certificate at level C2);

Spanish - Intermediate;

French - Intermediate.

AWARDS

Olympiads

2019: Brazilian Mathematics Olympiad for Public Schools (OBMEP) - **Bronze Medal**;

2019: Invited for participation in Brazilian Mathematics Olympiad (OBM).